DETERMINISTIC INVENTORY MODELING

Notation

c: unit purchase cost ($/unit)
K: fixed ordering cost ($/order)
i: annual inventory carrying rate ($/$/yr)
h=ic: unit annual inventory holding cost ($/unit/yr)
λ: demand rate (units/yr)
τ: constant lead time (yrs)
r: reorder level (units)

Cycle Length

$$T = \frac{q}{\lambda}$$

Single product: Retail/Procurement setting

$$C(q) = c\lambda + \frac{K\lambda}{q} + \frac{icq}{2}$$

$$q = \sqrt{\frac{2K\lambda}{h}}$$

Single product: Production setting

$$C(q) = c\lambda + \frac{K\lambda}{q} + \left(1 - \frac{\lambda}{P}\right)\frac{icq}{2}$$

$$q = \sqrt{\frac{2K\lambda}{ic\left(1 - \frac{\lambda}{P}\right)}}$$

Reorder level

Given the constant lead time of τ

Reorder level: $r = mod(\lambda\tau, q)$

Discount schemes

A discount scheme is characterized by a set of break points, indexed by $j = 0, \dots, r$, $b_0, b_1, \dots, b_r$, where

1. $b_0 < b_1 < \cdots b_r$ and
2. $b_0 = 0$

along with a set of unit prices $c_0, c_1, \dots, c_r$ where

$$c_0 > c_1 > \cdots > c_r$$

All-units discount scheme

<u>Step 1:</u> Evaluate the economic order quantity value for each price level c_j using

$$q_j = \sqrt{\frac{2K\lambda}{ic_j}} \text{ for } j = 1, \dots, r.$$

<u>Step 2:</u> For each order quantity evaluated in Step 1 and the corresponding lower and upper bounds for the price range, set feasible order quantities using:

1. If $q_j < b_j$ then $q_j^* = b_j$.
2. If $b_j \le q_j < b_{j+1}$ then $q_j^* = q_j$.
3. If $b_{j+1} < q_j$ then there is no candidate solution.

<u>Step 3:</u> Evaluate average annual total cost for each price level c_j using

$$C(q_j^*) = c_j\lambda + \frac{K\lambda}{q_j^*} + \frac{ic_jq_j^*}{2}$$

<u>Step 4:</u> Over all j, select order quantity with the lowest total cost.

Incremental discount scheme

<u>Step 1:</u> Evaluate V_j values at each price level c_j using

$V_0 = 0$

$V_j = c_0(b_1 - b_0) + c_1(b_2 - b_1) + \cdots + c_{j-1}(b_j - b_{j-1})$

<u>Step 2:</u> Evaluate EOQ values at each price level c_j using

$$q_j = \sqrt{\frac{2\lambda(K + V_j - b_jc_j)}{ic_i}} \text{ for } j = 1, \dots, r.$$

<u>Step 3:</u> For each order quantity evaluated in Step 1 and the corresponding lower and upper bounds for the price range, set feasible order quantities using

4. If $q_j < b_j$ then $q_j^* = b_j$.
5. If $b_j \le q_j < b_{j+1}$ then $q_j^* = q_j$.
6. If $b_{j+1} < q_j$ then $q_j^* = b_{j+1}$.

<u>Step 4:</u> Evaluate average annual total cost for each price level c_j and choose the one that yields the minimum average annual total cost using

$$C_j(q_i^*) = \frac{K\lambda}{q_i^*} + \left[V_j + (q_i^* - b_j)c_j\right]\frac{i}{2} + \left[V_j + (q_i^* - b_j)c_j\right]\frac{\lambda}{q_i^*}$$